

Vaughn Khouri

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EDUCATION

Massachusetts Institute of Technology (MIT)

Cambridge, MA

B.S. Mechanical Engineering, Concentration in Control, Instrumentation & Robotics (2A-6)

2025-2029

- Planned Minor in Materials Science and Engineering (Course 3)
- GPA: 5.0/5.0

EXPERIENCE

Dynamometer Lead, Inverter Boxen RE

Sept. 2025 – Present

MIT Motorsports FSAE Team

Cambridge, MA

- Led the design (Onshape) and fabrication (CNC lathe, mill) of a dynamometer platform to characterize motors and custom inverters under prolonged load. Sized for accurate torque measurements at a peak output power of 40 kW and 350 Hz driving frequency, verified via FEA (NX SimCenter). Wrote and presented several design reviews.
- Led the design (NX) and currently manufacturing MY26's Custom Inverter Box, housing 4 custom inverters, watercooled coldplates, and formed bus bars, reducing system volume by 4x compared to MY25.
- Aided fabrication process for custom sun, compound planet, and ring gears (CNC lathe, Wire EDM, waterjet).

Undergraduate Researcher

January 2026 – Present

Improbable AI Lab, MIT Computer Science & Artificial Intelligence Laboratory

Cambridge, MA

- Designing & manufacturing the next iteration of DexWrist, a novel 2-DOF spherical Parallel Kinematic Mechanism designed for dexterous manipulation tasks. Working under the guidance of Dr. Pulkit Agrawal and Martin Peticco.
- Integrated opposed, pre-loaded angular contact and adjustable linear ball bearings to eliminate backlash without overconstraint. Designed & machined testing jigs to verify lifespan of linear bearings under radial compression.

Mechatronics Engineer (Part-time)

November 2025 – Present

Alegria Neurotech

San Francisco, CA

- Owned all early stage hardware and electronics development, including:
 - * Re-engineered housings, power supplies, and acoustic lenses for head-mounted Butterfly ultrasound probes, used for real-time brain imaging, increasing contrast by 3x while maintaining real-time imaging.
 - * Sized and designed a custom blower-fan based cooling system with 3D-printed aluminum heatsinks for cooling skin-interface surfaces, reducing steady-state temperature by 60%.
 - * Designed and built individually fitted (MRI scan based) and universal fit adjustable headsets with 6-DOF lockable linkages for accurately and rigidly positioning imaging probes on participants.

Founding Engineer, Platform Development

May 2025 – Sept. 2025

Acre Robotics

Redwood City, CA

- Led hardware development from Seed stage to Series A, working closely with founders, CV, and Controls teams.
- Owned design, installation, and bring-up of custom hardware (Solidworks) and electronics (Jetson, ESP32, Stereolabs Zed, Ouster LiDAR), enabling the first autonomous driving tests on a full-size Case 75A tractor.
 - * Reverse-engineered the tractor ECU and other onboard auxiliary devices to interface with 10+ OEM sensors.
- Shipped a full-featured MVP with RTK GPS for stress testing on an operational farm in under 30 days.

Team Captain

Sept. 2021 – Jul. 2025

FIRST Robotics Competition Team 8033

Piedmont, CA

- Drove growth from founding season to a robust organization with 50+ students from 17+ high schools, placing in the top 16 worldwide (2024) and top 15 in California (2022–2025).
- Prototyped, designed, and iterated robot hardware (Onshape); set design requirements; led training and all-team meetings; created and documented design standards and workflows; and drove robot at 4 world championships.
- Designed (Onshape) and coordinated fabrication of a cost-effective, full-size (54' × 26') welded steel testing arena.

TECHNICAL SKILLS

CAD / CAM Tools: PTC Onshape, Siemens NX, SolidWorks, Fusion 360, Inventor, CATIA (including GD&T)

Machine Tools: Manual/CNC Mill, Lathe, MIG & TIG Welding, Waterjet / Fiber Laser, FDM / SLA / SLS 3D Printing, Sodick Wire Cut and Sinker EDM, All Common Wood / Metalworking Tools

Other: Conversational Mandarin (reading & writing), First Aid, outdoorsmanship (BSA Eagle Scout), Microsoldering & Basic PCB Design, Custom Battery Pack Design / Fabrication

More info and full project writeups available on my portfolio: <https://www.vkhouri.dev>